

Laporan Praktikum Jaringan Komputer IF-04-02

MODUL 10 IP

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Tujuan : Mahasiswa dapat menginvestigasi cara kerja protokol IP menggunakan Wireshark

Menangkap paket dari eksekusi traceroute >tracert gaia.cs.umass.edu

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Ryoo>tracert gaia.cs.umass.edu

Tracing route to gaia.cs.umass.edu [128.119.245.12]
over a maximum of 30 hops:

  1  16 ms  3 ms  3 ms  10.218.0.253
  2  19 ms  14 ms  10 ms  10.217.15.254
  3  25 ms  7 ms  4 ms  36.67.147.33
  4  34 ms  26 ms  *  180.240.190.101
  5  *  *  *  Request timed out.
  6  41 ms  24 ms  24 ms  180.240.204.8
  7  60 ms  26 ms  26 ms  be4126.ccr81.sin01.atlas.cogentco.com [154.18.3.249]
  8  197 ms  193 ms  191 ms  be9040.ccr41.lax04.atlas.cogentco.com [154.54.87.145]
  9  191 ms  192 ms  200 ms  be3271.ccr41.lax01.atlas.cogentco.com [154.54.42.101]
 10  204 ms  204 ms  203 ms  be2931.ccr31.phx01.atlas.cogentco.com [154.54.44.85]
 11  208 ms  209 ms  211 ms  be5471.ccr21.elp02.atlas.cogentco.com [154.54.166.57]
 12  222 ms  222 ms  221 ms  be3821.ccr31.dfw01.atlas.cogentco.com [154.54.165.25]
 13  236 ms  240 ms  235 ms  be5383.ccr31.oma02.atlas.cogentco.com [154.54.7.234]
 14  242 ms  243 ms  243 ms  be5214.ccr41.ord01.atlas.cogentco.com [154.54.165.134]
 15  248 ms  247 ms  248 ms  port-channel2717.ccr91.cle04.atlas.cogentco.com [154.54.6.222]
 16  250 ms  251 ms  252 ms  be8022.ccr21.alb02.atlas.cogentco.com [154.54.169.186]
 17  267 ms  259 ms  261 ms  be8027.ccr71.orh02.atlas.cogentco.com [154.54.169.213]
 18  262 ms  262 ms  264 ms  be2191.ccr71.orh01.atlas.cogentco.com [154.54.160.70]
 19  262 ms  264 ms  261 ms  38.32.33.83
 20  275 ms  262 ms  262 ms  69.16.0.9
 21  276 ms  261 ms  266 ms  69.16.3.0
 22  280 ms  265 ms  265 ms  core2-rt-et-4-3-0.gw.umass.edu [192.80.83.105]
 23  275 ms  272 ms  271 ms  n5-rt-1-1-et-10-0-0.gw.umass.edu [128.119.0.10]
 24  268 ms  269 ms  268 ms  128.119.77.235
 25  *  *  *  Request timed out.
 26  266 ms  267 ms  267 ms  cics-rt-xe-0-0-0.gw.umass.edu [128.119.3.32]
 27  *  *  *  Request timed out.
 28  266 ms  270 ms  265 ms  gaia.cs.umass.edu [128.119.245.12]

Trace complete.

C:\Users\Ryoo>
```

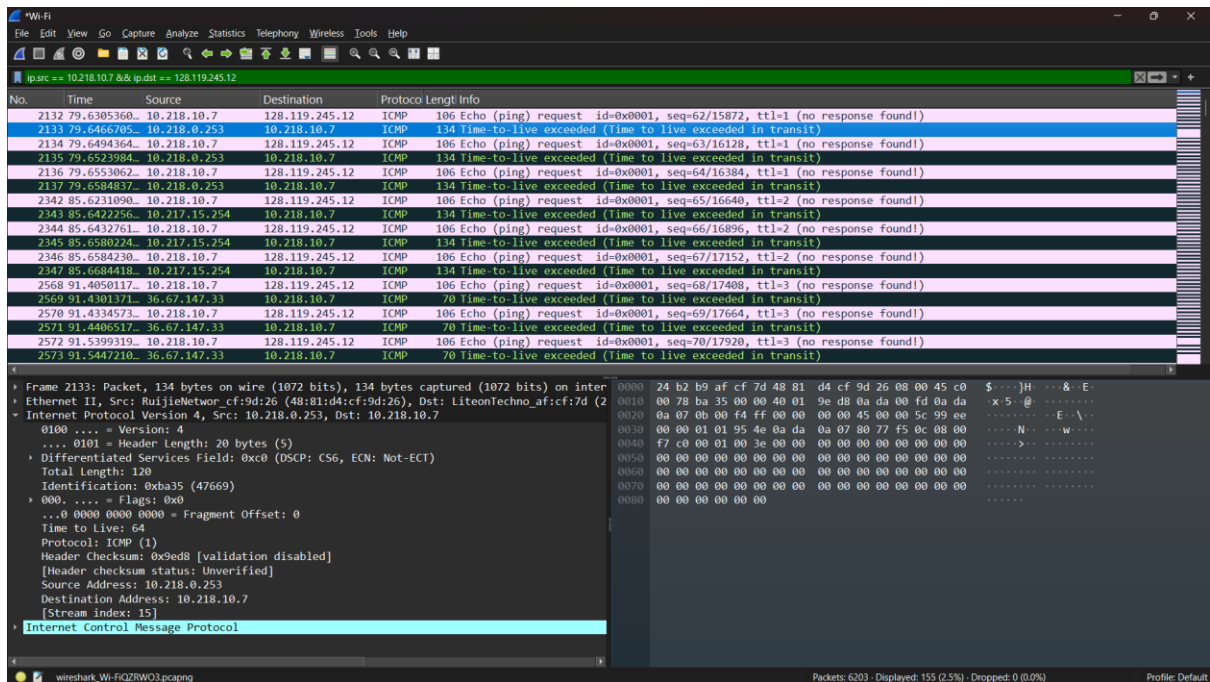
```
*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ipsrc == 10.218.10.7 && ipdst == 128.119.245.12

No. Time Source Destination Protocol Length Info
2132 79.6305360 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=62/15872, ttl=1 (no response found)
2133 79.6466705 10.218.0.253 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2134 79.6494364 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=63/16128, ttl=1 (no response found)
2135 79.6523984 10.218.0.253 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2136 79.6553062 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=64/16384, ttl=1 (no response found)
2137 79.6584837 10.218.0.253 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2342 85.6231090 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=65/16640, ttl=2 (no response found)
2343 85.6262256 10.217.15.254 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2344 85.6432761 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=66/16996, ttl=2 (no response found)
2345 85.6580224 10.217.15.254 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2346 85.6584230 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=67/17152, ttl=2 (no response found)
2347 85.6604418 10.217.15.254 10.218.10.7 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
2568 91.4050117 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=68/17408, ttl=3 (no response found)
2569 91.4301371 36.67.147.33 10.218.10.7 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2570 91.4334573 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=69/17664, ttl=3 (no response found)
2571 91.4405917 36.67.147.33 10.218.10.7 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2572 91.5393919 10.218.10.7 128.119.245.12 ICMP 106 Echo (ping) request id=0x0001, seq=70/17920, ttl=3 (no response found)
2573 91.5447210 36.67.147.33 10.218.10.7 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)

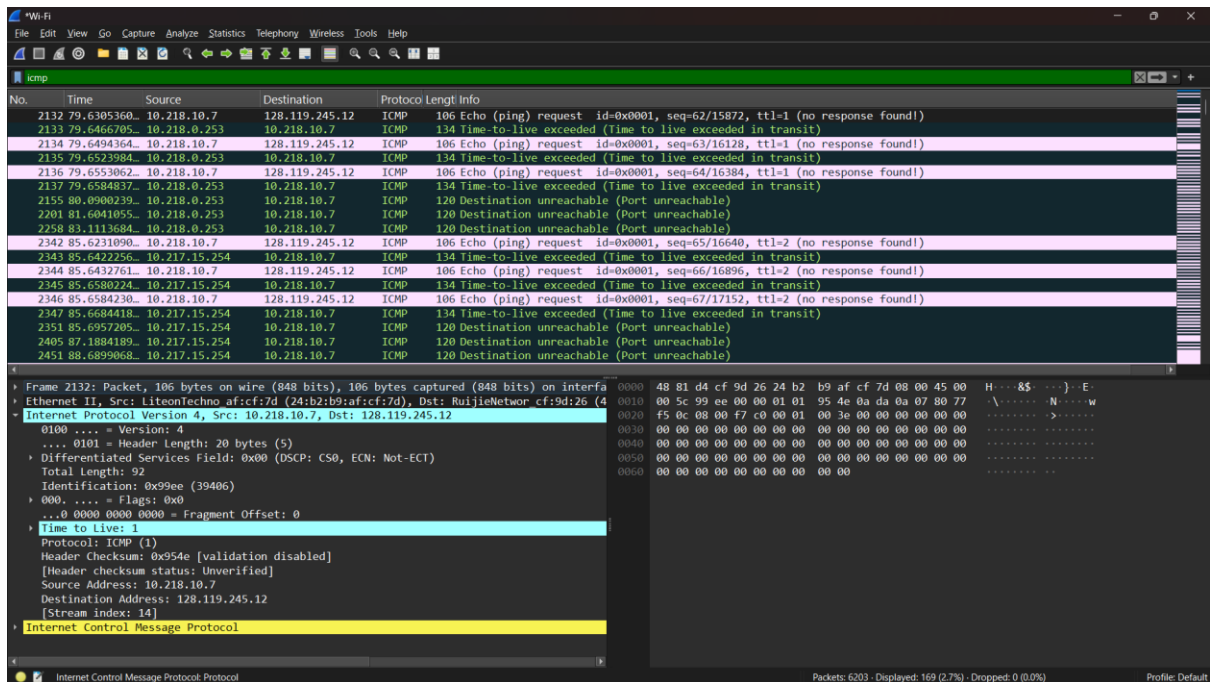
Frame 2132: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface 0
Ethernet II, Src: LiteonTechno af:cf:7d (24:b2:b9:af:cf:7d), Dst: RuijieNetwork cf:9d:26 (4:00:00:00:00:00)
Internet Protocol Version 4, Src: 10.218.10.7, Dst: 128.119.245.12
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 92
Identification: 0x99ee (39406)
0000 .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 1
Protocol: ICMP (1)
Header Checksum: 0x954e [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.218.10.7
Destination Address: 128.119.245.12
[Stream index: 14]
Internet Control Message Protocol

Packets: 6203 - Displayed: 155 (2.5%) - Dropped: 0 (0.0%) Profile: Default
```

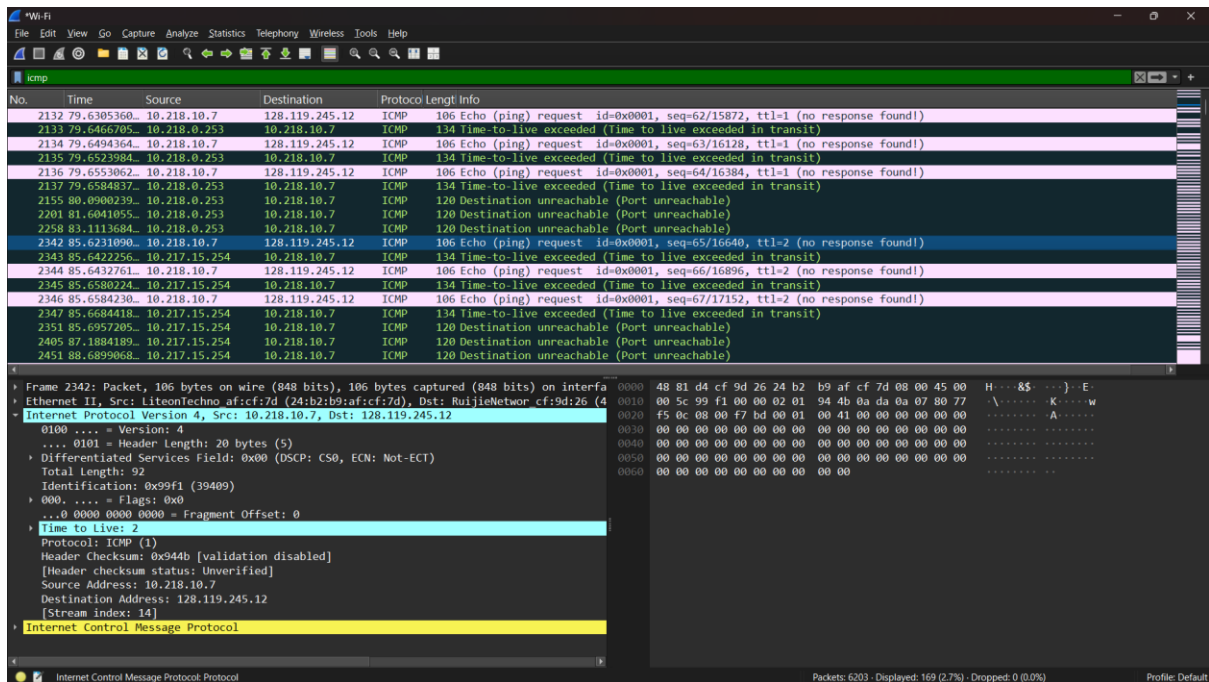


IPv4 Dasar

TTL 1:

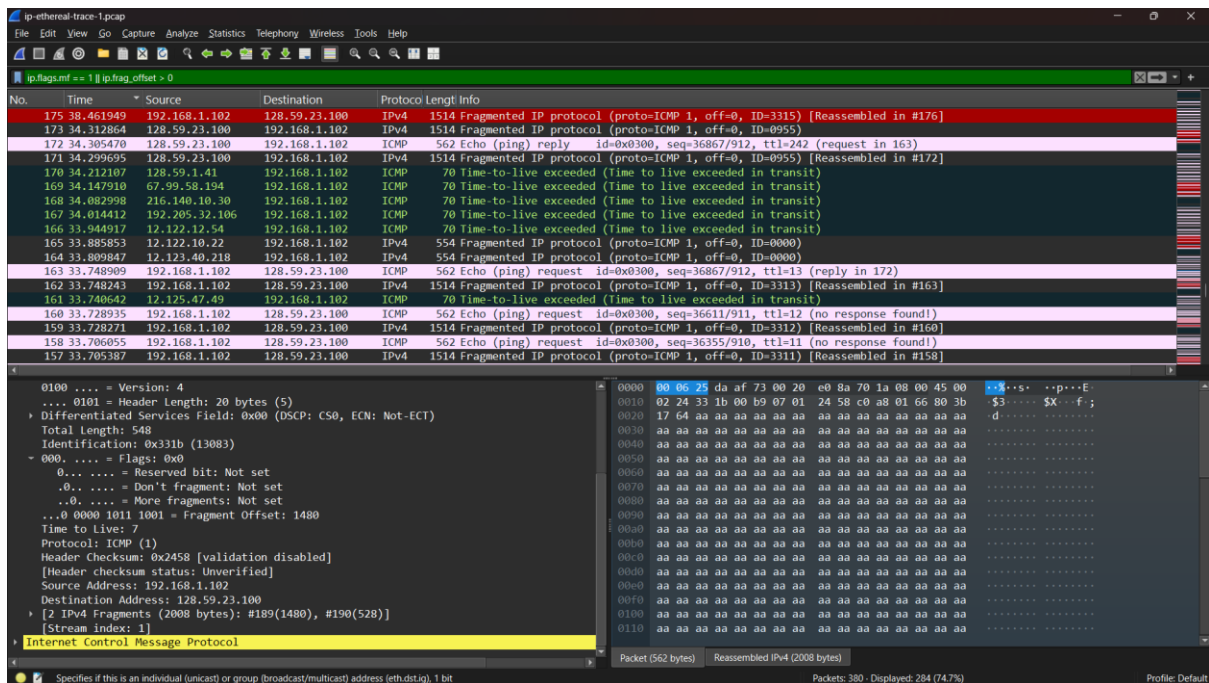


TTL 2:



Pada hasil capture Wireshark, terlihat bahwa komputer mengirim ICMP Echo Request dengan nilai TTL = 1 ke tujuan 128.119.245.12. Router pertama kemudian mengurangi nilai TTL menjadi 0 dan mengirimkan balasan ICMP “Time-To-Live Exceeded”. Hal ini menunjukkan bahwa traceroute berhasil mengidentifikasi hop pertama dalam jaringan.

Fregmentasi



Pada hasil analisis ditemukan fragmentasi IP. Hal ini ditunjukkan oleh adanya paket dengan nilai Fragment Offset lebih dari 0, yaitu sebesar 1480. Selain itu, nilai More Fragments (MF) pada paket tersebut adalah 0, yang menunjukkan bahwa paket tersebut merupakan fragment terakhir. Fragmentasi terjadi karena datagram yang dikirim melebihi ukuran Maximum

Transmission Unit (MTU), sehingga dipecah menjadi beberapa bagian. Hal ini diperkuat dengan adanya beberapa paket yang memiliki nilai Identification yang sama.

IPv6

Berdasarkan hasil analisis menggunakan Wireshark, ditemukan paket DNS yang berisi permintaan tipe A (IPv4), seperti pada domain youtube.com dan doubleclick.net. Namun, tidak ditemukan paket DNS dengan tipe AAAA (IPv6). Hal ini menunjukkan bahwa pada saat proses capture dilakukan, sistem hanya melakukan resolusi alamat IPv4 dan tidak melakukan permintaan alamat IPv6. Kemungkinan hal ini disebabkan oleh konfigurasi jaringan atau DNS server yang tidak mengutamakan penggunaan IPv6.

